Preconditioning for quantum linear solvers in PDE-constrained optimization

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Abstract

Quantum linear system solvers cost a number of operations proportional to an *effective* condition number. This number is the ratio of the normalization with which the matrix is embedded in a unitary circuit (its block encoding) to its smallest singular value. Preconditioning is therefore the main lever that numerical linear algebra has on quantum solvers. The rules differ from the classical ones, however:

- clustering of eigenvalues does not help;
- preconditioners must be applied by multiplication, not by inner solves;
- every factor of a product must be well scaled.

We develop preconditioners under these rules for the saddle-point systems of PDE-constrained optimization. For distributed control of the Poisson equation, we combine a Schöberl–Zulehner block preconditioner with an additive multilevel (BPX) factor. We apply it on both sides of the KKT matrix and factor the result as $\mathcal{G}^* \mathcal{J} \mathcal{G}$, where $\mathcal{J}$ is a single two-qubit reflection and $\mathcal{G}$ is well scaled. Building on the factorized multilevel block encodings of Deiml and Peterseim, which we formulate through quadrature points so that they cover Q_k and P_k elements, $\mathcal{G}$ has a block encoding whose normalization grows only like $\sqrt{\log h^{-1}}$. The effective condition number is then bounded independently of the mesh size. With level weights that we introduce, it is also bounded in the regularization parameter, up to a constant fixed by the mass matrix. For the time-periodic heat equation, a quantum Fourier transform in time decouples the problem into frequency-wise saddle-point problems of the same form. We prove a $\sqrt{3}$ bound uniform in all parameters, and the coupling matrix becomes a scaled reflection. Numerical experiments in one, two and three dimensions confirm the bounds. We also discuss when an asymptotic quantum advantage can and cannot be expected.

1 Introduction

Quantum linear system algorithms (QLSAs) prepare a quantum state proportional to the solution x of $Ax = b$. They use a number of calls to a circuit encoding A that scales like $\kappa \log(1/\varepsilon)$, where ε is the accuracy of the prepared solution state [5, 6, 9]. The size n of the system enters only through the circuit cost, which is typically polylogarithmic in n for structured problems. Discretized PDEs are the obvious application, but the condition number of a discretized differential operator grows with the resolution, e.g. like h^{-2} for second-order elliptic problems. Without preconditioning, the dependence on κ cancels the gain in n [15, 17].

Preconditioning for QLSAs has been studied:

- by sparse approximate inverses [4];
- by "fast inversion" of a dominant part of the matrix [25];
- by wavelet bases [2];

- for elliptic finite element discretizations, by Deiml and Peterseim [8]. Their work is the basis of our Section 3.

  – They write the stiffness matrix as $C_L^T(D_A \otimes I)C_L$, where C_L maps nodal values to gradients in an L^2-orthonormal basis of discontinuous Q_1 functions, and D_A holds the (possibly variable) diffusion coefficient.

  – They apply the BPX preconditioner as a generating system F, with FF^T the preconditioner, to this factor.

  – A commutative diagram (gradient on the coarse level, followed by an inclusion with orthonormal columns) gives a block encoding of the preconditioned factor whose normalization grows only linearly in the number of levels.

  – They introduce the effective condition number (subnormalization times condition number), and they ran small instances on quantum hardware.

  Related constructions are the finite-difference analogue [11] and the extension to reaction–diffusion problems $-\Delta u + cu$ with a bound uniform in c [27];

- empirically, for CFD matrices, comparing several preconditioners and ways of encoding them [16];

- for the Newton systems of quantum interior point methods [21], which become increasingly ill conditioned along the central path.

These works make clear that preconditioning for a quantum solver is a different problem from classical preconditioning (Section 2). Only a bound on the condition number counts, not the distribution of the eigenvalues. The preconditioner must be realized as a circuit that *multiplies*. And the normalizations of all factors enter the cost.

Gap. The saddle-point systems of PDE-constrained optimization have not been considered from this angle. Classically, they are the prototype for parameter-robust block preconditioners [20, 22, 24], which are robust in the mesh size h and the regularization parameter β. Quantum approaches to PDE-constrained optimization have so far been variational heuristics.

Contributions.

1. We formulate design principles for preconditioners of QLSAs in matrix terms, centred on the effective condition number κ_α (Section 2).

2. For distributed control of the Poisson equation (Section 3), we combine the Schöberl–Zulehner block preconditioner with a diagonally scaled BPX factor R, applied on both sides.

   - The resulting matrix factors as $\mathcal{G}^*\mathcal{J}\mathcal{G}$. Here $\mathcal{J}$ is a single two-qubit reflection, and $\mathcal{G}$ maps multilevel coefficients to values and gradients at quadrature points.
   - The block encoding of $\mathcal{G}$, with normalization $O(\sqrt{\log h^{-1}})$, builds on the commuting diagram of Deiml and Peterseim [8], extended to the mass term in [27]. We formulate it through quadrature points, with an *element-local isometry* (Lemma 3.6), so that it covers Q_k and P_k elements.
   - The effective condition number is bounded independently of h (Theorem 3.8). It holds for Q_k and P_k elements on nested meshes.

3. We introduce level weights that make the multilevel factor robust in β, up to the constant of the diagonally scaled mass matrix (Section 3.3).

4. For the time-periodic heat equation (Section 4), the QFT in time reduces the problem to frequency-wise problems of the Poisson type, with effective parameter $\beta_k = \beta/r_k^2$.
   - We prove a bound of $\sqrt{3}$ for the block preconditioner, uniform in h, τ, β and the frequency (Theorem 4.1).
   - The coupling matrix becomes a scaled reflection of norm at most one.
   - Neither the number of time steps nor the step size enters the effective condition number.
5. We verify all bounds and factorizations numerically in one, two and three dimensions (Section 6). We give an honest discussion of complexity: an asymptotic advantage over optimal classical solvers requires $d = 3$ for the elliptic problem. For the time-periodic problem it requires data that are rough in time (Section 7).

2 Quantum linear algebra in matrix terms

We describe the computational model in the language of numerical linear algebra (see [9,14] for details).

States and outputs. A unit vector $v \in \mathbb{C}^{2^k}$ is stored in k qubits. Its entries cannot be read out. Repeated measurement gives estimates of quantities v^*Ov for suitable Hermitian O, with additive error η. This costs $O(\eta^{-2})$ repetitions, or $O(\eta^{-1})$ with amplitude estimation [1]. The output of a quantum solver is therefore a small number of scalar *quantities of interest* (QoIs), such as $\langle w, y\rangle_M$ or the value of the objective, never the vector x.

Block encodings. A matrix A with $\|A\| \le \alpha$ is *block encoded* with *subnormalization* α if a unitary circuit U satisfies

$$U = \begin{bmatrix} A/\alpha & * \\ * & * \end{bmatrix}.$$

Every block of a unitary matrix has norm at most one, so $\alpha \ge \|A\|$ for *every* block encoding. We call $\alpha = \|A\|$ the *ideal subnormalization*. It is always attainable in principle, by a unitary dilation, but that dilation is a dense matrix without an efficient circuit. Efficient circuits are built from structure, and they usually pay more than $\|A\|$. How much more depends on the construction; examples are in the supplement (`nla_for_quantum.tex`). Applying U to $[v; 0]$ and keeping the first block (postselection) yields $Av/\|Av\|$. This succeeds with probability $\|Av\|^2/(\alpha\|v\|)^2$.

A circuit is a product of unitary matrices (gates), each acting on a few of the tensor factors $\mathbb{C}^2$. For small problems, the whole product can be formed as an explicit matrix and checked. As an example, let S be the cyclic shift (a permutation matrix) and $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Then

$$U = (H \otimes I)\begin{bmatrix} I & 0 \\ 0 & -S \end{bmatrix}(H \otimes I) = \frac{1}{2}\begin{bmatrix} I - S & I + S \\ I + S & I - S \end{bmatrix}$$

is unitary, and it block encodes the difference matrix $I - S$ with $\alpha = 2$. This is the simplest instance of a linear combination of unitaries (LCU): H prepares the weights, and the middle factor selects I or $-S$ according to one extra qubit. Rectangular matrices are padded with zeros to square matrices of power-of-two size. Block encodings combine as follows (subnormalizations in brackets):

- products AB $[\alpha_A\alpha_B]$;

- Kronecker products $A \otimes B$ $[\alpha_A \alpha_B]$;
- linear combinations $\sum_j c_j A_j$ $[\sum_j |c_j| \alpha_j]$;
- stacking $[A_1; \ldots; A_k]$ $[(\sum_j \alpha_j^2)^{1/2}]$: prepare an index register with amplitudes $\propto \alpha_j$ and apply A_j controlled on index j;
- concatenation $[A_1, \ldots, A_k]$ $[(\sum_j \alpha_j^2)^{1/2} \le \sqrt{k} \max_j \alpha_j]$;
- sparse matrices with s_r nonzeros per row and s_c per column, whose entries are computable by a circuit $[\sqrt{s_r s_c} \max_{ij} |a_{ij}|]$ [9].

Products need a fresh ancilla per factor; reusing one ancilla brings discarded components back into the leading block. Explicit unitary realizations of all rules, with derivations of the subnormalizations and numerical checks, are given in the supplement *Block encodings for numerical linear algebraists* (`nla_for_quantum.tex`). In a product, the subnormalization can be far larger than the norm: $\alpha_A \alpha_B \ge \|A\|\|B\| \gg \|AB\|$ is possible.

Solving. Let $\mathcal{B}$ be Hermitian and block encoded with subnormalization α, and let its nonzero eigenvalues satisfy $|\lambda| \ge \sigma_{\min}$. A quantum solver for $\mathcal{B}z = b$ outputs a unit vector $\tilde{z}$ (a quantum state). Its *accuracy* ε is measured in the Euclidean norm against the normalized minimum-norm solution:

$$\left\| \tilde{z} - \frac{\mathcal{B}^+ b}{\|\mathcal{B}^+ b\|} \right\| \le \varepsilon. \tag{1}$$

Throughout, ε denotes this solver accuracy. The additive error of a measured scalar is η (above). The target accuracy of a quantity of interest is tol (Section 7). Quantum singular value transformation (QSVT) [9] applies an odd polynomial $p(\mathcal{B}/\alpha)$ of degree D using D calls to the block encoding.

Why odd. QSVT alternates calls to U and U^*. It acts on the singular values $|\lambda_i|$ of $\mathcal{B} = Q\Lambda Q^*$, and D calls give a polynomial p of degree D and parity D (as in $(\mathcal{B}^*\mathcal{B})^k$ versus $\mathcal{B}(\mathcal{B}^*\mathcal{B})^k$). The signs of the eigenvalues enter only through their D-th power:

$$Q \operatorname{diag}\left(\operatorname{sign}(\lambda_i)^D\right) p(|\Lambda|)\, Q^*.$$

For odd D this is $Qp(\Lambda)Q^* = p(\mathcal{B})$. For even D the signs are lost, so an even p cannot approximate $1/\lambda$ on an indefinite spectrum.

Solving means evaluating a polynomial. A QSVT solver does not iterate. It evaluates the matrix function $p(\mathcal{B}/\alpha)$ for a fixed odd polynomial with

$$p(t) \approx \frac{\delta}{t} \ \text{ on } [\delta, 1], \qquad |p| \le 1 \ \text{ on } [-1, 1], \qquad \delta = \frac{\sigma_{\min}}{\alpha}.$$

The bound $|p| \le 1$ is needed because $p(\mathcal{B}/\alpha)$ must again be a block of a unitary. Since $p(0) = 0$, the result is the scaled pseudoinverse $\delta\, \mathcal{B}^+$.

- In classical terms this is Chebyshev semi-iteration whose degree is fixed in advance, namely $O(\kappa_\alpha \log(\kappa_\alpha/\varepsilon))$. The $\log(\kappa_\alpha/\varepsilon)$ arises because p must approximate δ/t to about $\varepsilon/\kappa_\alpha$ for the normalized output to satisfy (1). The polynomial depends only on the interval, not on b or the actual spectrum. Unlike CG or MINRES, it cannot adapt.
- Postselecting $\delta\mathcal{B}^+ b$ succeeds with probability

$$\frac{\|\delta\mathcal{B}^+ b\|^2}{\|b\|^2} \ \ge \ \frac{1}{\kappa_\alpha^2}.$$

  Amplitude amplification then needs at most κ_α rounds.

Plain QSVT inversion therefore costs $O(\kappa_\alpha^2 \log(\kappa_\alpha/\varepsilon))$ calls. Variable-time amplitude amplification [5] or the discrete adiabatic method [6] achieve the optimal

$$O\big(\kappa_\alpha \log(1/\varepsilon)\big), \qquad \kappa_\alpha := \frac{\alpha}{\sigma_{\min}} \;\geq\; \kappa(\mathcal{B}). \tag{2}$$

In both cases the cost is governed by κ_α, the *effective condition number* in the terminology of Deiml and Peterseim [8, Def. 4.1].

What preconditioning means. If solving means evaluating a polynomial, then preconditioning means *choosing the matrix in which the polynomial is evaluated*, so that a polynomial of low degree suffices.

Why this is the only lever. The polynomial has to approximate δ/t on $[\delta, 1]$ while staying bounded by one on $[-1, 1]$, where $\delta = \sigma_{\min}/\alpha = 1/\kappa_\alpha$. This needs degree about $\kappa_\alpha \log(1/\varepsilon)$, and every degree is one call to the block encoding. Solving is cheap only if δ is not small: the *scaled* spectrum of the encoded matrix must stay away from zero.

How it is done. Let $\mathcal{A}x = f$ be the system and let $\widetilde{R}$ be a factor of the preconditioner, $\widetilde{R}\widetilde{R}^T \approx P^{-1}$. Instead of evaluating a polynomial in $\mathcal{A}/\alpha_{\mathcal{A}}$, we evaluate one in the preconditioned matrix $\mathcal{B} = \widetilde{R}^T \mathcal{A} \widetilde{R}$ and map back:

$$\mathcal{A}^{-1} = \widetilde{R}\, \mathcal{B}^{+} \widetilde{R}^T \;\approx\; \frac{1}{\sigma_{\min}(\mathcal{B})}\, \widetilde{R}\, p\big(\mathcal{B}/\alpha_{\mathcal{B}}\big)\, \widetilde{R}^T,$$

with $p \approx \delta/t$ as above (Theorem 3.5 justifies the first equality). The degree drops from about $\kappa_\alpha(\mathcal{A}) \log(1/\varepsilon)$ to about $\kappa_\alpha(\mathcal{B}) \log(1/\varepsilon)$. Both are effective condition numbers, because the cost is always set by the encoded matrix. For $\mathcal{A}$ we only use $\kappa_\alpha(\mathcal{A}) \geq \kappa(\mathcal{A})$, with equality for an ideal encoding; this is the best case for the unpreconditioned solver. For $\mathcal{B}$ the gap between $\kappa_\alpha(\mathcal{B})$ and $\kappa(\mathcal{B})$ is exactly what the encoding lever below controls, so $\kappa(\mathcal{B})$ alone would be misleading. For the Poisson control problem of Section 3 with $d = 1$ and $h = 2^{-8}$, this is $\kappa_\alpha(\mathcal{A}) \geq \kappa(\mathcal{A}) \approx 2.6 \cdot 10^4$ against $\kappa_\alpha(\mathcal{B}) \approx 5$–$10$. The latter assumes an ideal encoding of the factor G of Section 3. There $\mathcal{B} = \mathcal{G}^T \mathcal{J} \mathcal{G}$ with $\mathcal{G} = I_2 \otimes G$, and G maps multilevel coefficients to values and gradients at quadrature points. The middle factor $\mathcal{J}$ costs nothing, so $\alpha_{\mathcal{B}} = \alpha_G^2$, which is $\|G\|^2$ in the ideal case. The value for the explicit construction is still to be measured (Section 3).

Two levers, one classical and one quantum. The ratio $\delta = \sigma_{\min}(\mathcal{B})/\alpha_{\mathcal{B}}$ can be spoiled from either side.

1. *Spectral (classical).* $\sigma_{\min}(\mathcal{B})$ must be comparable to $\|\mathcal{B}\|$. This is ordinary spectral equivalence of P and $\mathcal{A}$, i.e. eigenvalue inclusion intervals. In Section 3 it is provided by the Schöberl–Zulehner block structure combined with BPX.

2. *Encoding (quantum).* $\alpha_{\mathcal{B}}$ must be comparable to $\|\mathcal{B}\|$. This fails if $\mathcal{B}$ is encoded as the product $\widetilde{R}^T \cdot \mathcal{A} \cdot \widetilde{R}$: then $\alpha_{\mathcal{B}} \gtrsim \|R\|^2 \|\mathcal{A}\|$, which is of the order of $\kappa(\mathcal{A})$ (Section 3, (10)), and lever 1 is undone. It succeeds if $\mathcal{B}$ is written as a product of factors that are each encoded close to their own norm, and whose norms multiply to about $\|\mathcal{B}\|$ (no cancellation between the factors). A Gram factorization $F^T F$ has the second property automatically, since $\|F^T F\| = \|F\|^2$. Our $\mathcal{B} = \mathcal{G}^T \mathcal{J} \mathcal{G}$ (Section 3) is of this type, with a reflection $\mathcal{J}$ in the middle to carry the indefiniteness.

Comparison with classical preconditioning.

- *Where the preconditioner sits.* Classically, P is applied once per iteration, usually by an inner solve such as a multigrid cycle. Here the preconditioner is built into the matrix that is encoded, so every call of the circuit already multiplies by the preconditioned operator. There are no inner solves: P^{-1} must be available as a multiplication by a factor $\widetilde{R}$ (principle (D3) below).

- *What is optimized.* A Krylov method chooses its polynomial adaptively, from the actual spectrum and the right-hand side, so clustering of eigenvalues helps. Here the polynomial is fixed in advance by the interval $[\delta, 1]$, so only the interval counts (principle (D1)).
- κ *versus* $\sqrt{\kappa}$. For SPD systems, Chebyshev iteration or CG need about $\sqrt{\kappa}$ steps, because they minimize a residual polynomial normalized by $p(0) = 1$, a relative error measure. QSVT must approximate the inverse itself uniformly on $[\delta, 1]$, while staying bounded by one on $[-1, 1]$. This needs degree of order κ_α. The quantum solver therefore depends more strongly on the condition number than CG does, which makes preconditioning *more* important, not less.
- *The ends of the product.* In practice $\widetilde{R}$ and $\widetilde{R}^T$ are not applied separately. The right-hand side is prepared as $\widetilde{R}^T f = \mathcal{G}^T g$, and outputs are read through $\mathcal{G}$ (principle (D5)).

Design principles. Equation (2) changes what a good preconditioner is.

(D1) *Only κ_α counts.* QSVT does not adapt to the spectrum, unlike MINRES. Three distinct eigenvalues help MINRES but not a QLSA. What we need are inclusion intervals $[-b, -a] \cup [a, b]$ with b/a independent of the parameters, i.e. exactly the robustness results of [20, 24].

(D2) *Subnormalizations count.* The preconditioned matrix must be encoded with α close to its norm. Multiplying separately encoded factors such as $P^{-1/2}$, $\mathcal{A}$ and $P^{-1/2}$ fails. This is why [8] precondition a *factor* of the stiffness matrix rather than the matrix itself; see also [16].

(D3) *Preconditioners multiply.* A classical inner solve (e.g. a multigrid cycle) has to become a circuit that multiplies by P^{-1} or by a factor of it. Additive multilevel methods [3, 26] are sums of products of structured sparse matrices and fit this model. Multiplicative multigrid does not.

(D4) *Kernels are harmless.* QSVT inverts on the range, so redundant generating systems, such as multilevel ones, are allowed. This is used in [8] for the BPX generating system.

(D5) *Organize for QoIs.* Right-hand sides and outputs should be accessible without applying further badly scaled operators.

3 Distributed control of the Poisson equation

3.1 Problem and discretization

Let $\Omega = (0, 1)^d$, $d \leq 3$, with desired state $\hat{y} \in L^2(\Omega)$ and regularization parameter $\beta > 0$. We solve

$$\min_{y,u} \ \tfrac{1}{2}\|y - \hat{y}\|^2_{L^2(\Omega)} + \tfrac{\beta}{2}\|u\|^2_{L^2(\Omega)} \quad \text{s.t.} \quad -\Delta y = u \text{ in } \Omega, \quad y = 0 \text{ on } \partial\Omega. \tag{3}$$

Grid and matrices. We use Q_1 elements on a uniform grid with $h = 2^{-L}$. There are 2^L elements and $2^L - 1$ interior nodes per direction, so $n = (2^L - 1)^d$ unknowns. In 1D, $M_1 = \frac{h}{6}\,\mathrm{tridiag}(1, 4, 1)$ and $K_1 = \frac{1}{h}\,\mathrm{tridiag}(-1, 2, -1)$. In d dimensions the matrices are Kronecker products, e.g. $M = M_1 \otimes M_1$ and $K = K_1 \otimes M_1 + M_1 \otimes K_1$ for $d = 2$. This structure matches the tensor-product structure of a qubit register.

Factorizations through quadrature points. Let $\xi_{1,2} = \frac{1}{2} \mp \frac{1}{2\sqrt{3}}$ be the two Gauss points on $[0,1]$. Let $F_0, F_1 \in \mathbb{R}^{2^{L+1} \times (2^L - 1)}$ map nodal values to $\sqrt{h/2}$ times the values, respectively derivatives, of the finite element function at the Gauss points of all elements. Two-point Gauss quadrature is exact for the integrands involved, so

$$M_1 = F_0^T F_0, \qquad K_1 = F_1^T F_1. \tag{4}$$

In d dimensions let $F_{\text{val}} = F_0^{\otimes d}$ and let F_∇ stack the factors $F^{(i)} = F_0 \otimes \cdots \otimes F_1 \otimes \cdots \otimes F_0$ (F_1 in position i). Then $M = F_{\text{val}}^T F_{\text{val}}$ and $K = F_\nabla^T F_\nabla$. The factors have $2^{d(L+1)}$ rows, 2^d nonzeros per row and at most 4^d per column.

This is the quadrature-point version of the factorization of Deiml and Peterseim [8, (6.1)], which uses an L^2-orthonormal (Legendre) basis of discontinuous Q_1 functions on each element instead of weighted values at Gauss points. For Q_1 the two are related by an orthogonal 2×2 map per element and direction, so the statements are equivalent. The quadrature form includes the mass matrix, as in [27], and extends directly to Q_k and P_k elements (Remark 3.1).

Remark 3.1 (Other elements). The construction is not specific to Q_1. Consider a nested family of conforming Lagrange spaces: Q_k on Cartesian grids, or P_k on uniformly red-refined, possibly unstructured simplicial meshes. Take a quadrature rule with positive weights that is exact for squares of the local polynomial space.

- The factorizations $M = F_{\text{val}}^T F_{\text{val}}$ and $K = F_\nabla^T F_\nabla$ are exact.
- The quadrature points are unisolvent, since $0 < \int p^2 = \sum_q w_q p(x_q)^2$.
- Lemma 3.6 below holds with an element-local isometry, the same for values and gradients, because derivatives stay in the local space.
- For Q_k the Kronecker form $M = M_1 \otimes M_1$, $K = K_1 \otimes M_1 + M_1 \otimes K_1$ ($d = 2$) holds for every k. The nodal basis functions are products $\phi_i(x)\phi_j(y)$ of 1D degree-k basis functions, so the integrals factor; only M_1 and K_1 change. For Q_2 they are 2-periodic (vertex and midpoint nodes alternate), with $2^{L+1} - 1$ unknowns per direction. Serendipity elements and P_k on simplices are not tensor products.
- Row counts that are not powers of two are zero-padded.
- Variable coefficients enter as one diagonal factor.

Mass lumping, curved elements and non-nested hierarchies keep the factorization only up to spectral equivalence. Unstructured meshes lose the Kronecker structure of the factors, which then need sparse-access oracles. We verified these statements numerically for Q_2 and Q_3 (1D and 2D) and for P_1 and P_2 on triangles. All factorization, commuting and isometry errors are below $3.3 \cdot 10^{-14}$.

Discrete problem and KKT system. Let $b = F_{\text{val}}^T \hat{y}_q$, where $\hat{y}_q$ holds the weighted values of $\hat{y}$ at the Gauss points. We minimize $\frac{1}{2} y^T M y - b^T y + \frac{\beta}{2} u^T M u$ subject to $Ky = Mu$. The optimality conditions are

$$\begin{bmatrix} M & 0 & K \\ 0 & \beta M & -M \\ K & -M & 0 \end{bmatrix} \begin{bmatrix} y \\ u \\ p \end{bmatrix} = \begin{bmatrix} b \\ 0 \\ 0 \end{bmatrix}. \tag{5}$$

We eliminate $u = p/\beta$, set $q = p/\sqrt{\beta}$ and scale the second equation. This gives

$$\mathcal{A}x = f, \qquad \mathcal{A} = \begin{bmatrix} M & \sqrt{\beta} K \\ \sqrt{\beta} K & -M \end{bmatrix}, \qquad x = \begin{bmatrix} y \\ q \end{bmatrix}, \quad f = \begin{bmatrix} b \\ 0 \end{bmatrix}, \tag{6}$$

with $u = q/\sqrt{\beta}$. The matrix $\mathcal{A}$ is symmetric and indefinite, with $\kappa(\mathcal{A}) = O(1 + \sqrt{\beta} h^{-2})$ (Table 1). Classically, MINRES with a robust block preconditioner and multigrid solves (6) in $O(n)$ operations [20, 22, 24].

3.2 A block preconditioner with a single elliptic operator

Let $A_\beta := M + \sqrt{\beta}K$, the stiffness matrix of $I - \sqrt{\beta}\Delta$.

Theorem 3.2 ([24]). *For $P = \mathrm{blkdiag}(A_\beta, A_\beta)$, $\mathrm{spec}(P^{-1}\mathcal{A}) \subset [-1, -1/\sqrt{2}] \cup [1/\sqrt{2}, 1]$ for all h and β.*

Proof. With $X^T M X = I$ and $X^T K X = \mathrm{diag}(\lambda_j)$, the pencil $(\mathcal{A}, P)$ decouples into 2×2 pencils $\Big(\begin{bmatrix} 1 & \mu_j \\ \mu_j & -1 \end{bmatrix}, (1+\mu_j)I \Big)$ with $\mu_j = \sqrt{\beta}\lambda_j > 0$. Their eigenvalues are $\pm\sqrt{1+\mu_j^2}/(1+\mu_j) \in \pm[1/\sqrt{2}, 1]$. □

Remark 3.3. In the Pearson–Wathen approximation of the Schur complement of (5) [20], $\widehat{S} = (K + M/\sqrt{\beta})M^{-1}(K + M/\sqrt{\beta})$, the operator to invert is again $K + M/\sqrt{\beta} = A_\beta/\sqrt{\beta}$. We use the form of Theorem 3.2 because it avoids M^{-1} and has the sharper interval.

3.3 A multilevel factor and level weights

For levels $\ell = 1, \ldots, L$ with $h_\ell = 2^{-\ell}$, let $P_\ell \in \mathbb{R}^{n \times n_\ell}$ be the prolongation from level ℓ to level L ($P_L = I$). The level matrices $A_{\beta,\ell}$ have constant diagonal on a uniform grid:

$$\mathrm{diag}(A_{\beta,\ell}) = \delta_\ell I, \qquad \delta_\ell = \big(\tfrac{2h_\ell}{3}\big)^d + \sqrt{\beta}\,\tfrac{2d}{h_\ell}\big(\tfrac{2h_\ell}{3}\big)^{d-1}.$$

Following [8, Sec. 5], we use BPX in factored form, as a generating system R with $C = RR^T$. We define the weighted, diagonally scaled BPX factor

$$R = \big[\, \omega_1 P_1,\ \ldots,\ \omega_L P_L \,\big], \qquad C = RR^T = \sum_{\ell=1}^{L} \omega_\ell^2 P_\ell P_\ell^T, \qquad \omega_\ell^2 = \frac{\theta_\ell}{\delta_\ell}, \tag{7}$$

with level weights

$$\theta_\ell = \min\big(1, \sqrt{\beta}\, h_\ell^{-2}\big)^s, \qquad s \in \{0, 1\}. \tag{8}$$

The choice $s = 0$ is plain BPX. Since $P_L = I$, R has full row rank.

Assumption 3.4. $\mathrm{spec}(CA_\beta) \subset [c_0, c_1]$, *with c_1/c_0 bounded independently of h and, ideally, of β.*

For fixed β and $s = 0$, this is the classical BPX theory for H^1-elliptic problems [3, 19, 26]. Robustness in β concerns the singularly perturbed operator $I - \sqrt{\beta}\Delta$.

- *Plain BPX* is robust in h but degrades as $\beta \to 0$ (Section 6): coarse levels on which the mass term dominates ($\sqrt{\beta}h_\ell^{-2} < 1$) all count the same smooth modes.
- *The weights* (8) with $s = 1$ damp these levels. They keep $\kappa(CA_\beta)$ small in all our tests, except when the grid does not resolve β at all ($h > \beta^{1/4}$). In that case $A_\beta \approx M$, and $\kappa(CA_\beta)$ approaches the constant 3^d of the Jacobi-scaled Q_1 mass matrix.
- *Truncation does not work.* Simply removing the mass-dominated levels makes things worse at intermediate β.

[TODO: Prove Assumption 3.4 for $s = 1$ with β-independent constants (robust additive multilevel methods for $I - \mu^2\Delta$); a better mass factor than diagonal scaling would lower the 3^d floor.]

3.4 The two-sided system and its factorization

Let $\widetilde{R} = I_2 \otimes R$ and

$$\mathcal{B} := \widetilde{R}^T \mathcal{A} \widetilde{R} = \begin{bmatrix} R^T M R & \sqrt{\beta} R^T K R \\ \sqrt{\beta} R^T K R & -R^T M R \end{bmatrix}. \tag{9}$$

Theorem 3.5. *Under Assumption 3.4:*

1. *$\mathcal{A}x = f$ if and only if $x = \widetilde{R}z$ for a solution z of the consistent system $\mathcal{B}z = \widetilde{R}^T f$. In particular $x = \widetilde{R}\mathcal{B}^+\widetilde{R}^T f$.*

2. *The nonzero eigenvalues of $\mathcal{B}$ lie in $[-c_1, -c_0/\sqrt{2}] \cup [c_0/\sqrt{2}, c_1]$.*

Proof. (1) $\widetilde{R}$ is surjective, so $\widetilde{R}^T$ is injective, and $\mathrm{range}(\mathcal{B}) = \widetilde{R}^T \mathcal{A} \mathbb{R}^{2n} = \mathrm{range}(\widetilde{R}^T)$. (2) The nonzero eigenvalues of $\widetilde{R}^T \mathcal{A} \widetilde{R}$ are those of $\widetilde{C}^{1/2} \mathcal{A} \widetilde{C}^{1/2} = S(P^{-1/2} \mathcal{A} P^{-1/2}) S^T$, where $\widetilde{C} = I_2 \otimes C$ and $S = \widetilde{C}^{1/2} P^{1/2}$. By Ostrowski's theorem [10, Thm. 4.5.9], each one equals $\theta\lambda$, where λ is an eigenvalue as in Theorem 3.2 and $\theta \in \mathrm{spec}(SS^T) = \mathrm{spec}(I_2 \otimes C A_\beta) \subset [c_0, c_1]$. □

Factorization with a reflection. *Why a factorization is needed.* Classically, a preconditioned operator $\widetilde{R}^T \mathcal{A} \widetilde{R}$ is applied factor by factor, and the norms of the factors are irrelevant. A Krylov method sees only the spectrum of the product. On a quantum computer, a product of block encodings encodes $\widetilde{R}^T \mathcal{A} \widetilde{R} / (\alpha_R^2 \alpha_{\mathcal{A}})$, because every factor is applied with a loss of amplitude. Its subnormalization satisfies

$$\alpha \;\geq\; \|R\|^2 \, \|\mathcal{A}\| \;=\; \|C\| \, \|\mathcal{A}\| \;\geq\; c_0 \|A_\beta^{-1}\| \cdot \tfrac{1}{2} \|A_\beta\| \;=\; \tfrac{c_0}{2} \, \kappa(A_\beta). \tag{10}$$

The steps are as follows.

- Each factor X has $\alpha_X \geq \|X\|$, since X/α_X is a block of a unitary matrix, and the α's multiply.
- $\|R\|^2 = \|RR^T\| = \|C\| \geq c_0 \|A_\beta^{-1}\|$ by Assumption 3.4.
- $\|A_\beta\| \leq \|M\| + \sqrt{\beta}\|K\| \leq 2\|\mathcal{A}\|$, because M and $\sqrt{\beta}K$ are blocks of $\mathcal{A}$.

Since $\sigma_{\min}(\mathcal{B}) \leq \|\mathcal{B}\| \leq c_1$, the effective condition number of the product encoding is at least $\frac{c_0}{2c_1} \kappa(A_\beta)$, with $\kappa(A_\beta) \sim 1 + \sqrt{\beta} h^{-2}$. Numerically this is close to $\kappa(\mathcal{A})$. For example, for $d = 1$, $L = 8$, $\beta = 1$: $\|R\|^2 \|\mathcal{A}\| = 3.0 \cdot 10^4$, $\kappa(\mathcal{A}) = 2.6 \cdot 10^4$, while the factored form below gives $\|G\|^2 / \sigma_{\min}(\mathcal{B}) = 5.0$. QSVT then has to invert on $[\sigma_{\min}/\alpha, 1]$, with degree $\sim \alpha/\sigma_{\min}$. It is non-adaptive: it behaves like Chebyshev semi-iteration with the upper eigenvalue bound overestimated as α instead of $\lambda_{\max}$. So the cost would be of the order of $\kappa(\mathcal{A})$, as without preconditioning. Krylov methods do not suffer from this, because they adapt to the actual spectrum.

What a good factorization needs. We look for a factorization $\mathcal{B} = F_1 F_2 \cdots F_m$ with two properties.

(W1) *Tight factors:* each factor has an efficient block encoding with $\alpha(F_i) \approx \|F_i\|$.

(W2) *No cancellation:* $\prod_i \|F_i\| \approx \|\mathcal{B}\|$. That is, the vectors on which one factor is large are not damped by its neighbours.

Then $\alpha_{\mathcal{B}} = \prod_i \alpha(F_i) \approx \|\mathcal{B}\|$. The naive product $\widetilde{R}^T \cdot \mathcal{A} \cdot \widetilde{R}$ satisfies (W1) but violates (W2): R is large on smooth vectors, where $\mathcal{A}$ is small, and $\|R\|^2 \|\mathcal{A}\|$ exceeds $\|\mathcal{B}\|$ by a factor of order $\kappa(A_\beta)$.

- *Gram factorizations satisfy (W2) automatically,* because $\|F^T F\| = \|F\|^2$ exactly. This is why we factor through $G = [F_{\mathrm{val}} R;\, \beta^{1/4} F_\nabla R]$, which has $G^T G = R^T A_\beta R$.

- *An indefinite $\mathcal{B}$ needs a reflection in the middle* (next paragraph). (W2) then holds up to a bounded factor: $\|\mathcal{G}\|^2\|\mathcal{J}\| = \|G\|^2 \le c_1$, while $\|\mathcal{B}\| \ge c_0/\sqrt{2}$ by Theorem 3.5, so $\prod_i \|F_i\|/\|\mathcal{B}\| \le \sqrt{2}\, c_1/c_0$.
- *(W1)* is the subject of Section 3.5 (Lemma 3.6 and Proposition 3.7).

Why a reflection in the middle. $\mathcal{B}$ is indefinite, so it cannot be written as G^TG. Let

$$G = \begin{bmatrix} F_{\text{val}}R \\ \beta^{1/4}F_\nabla R \end{bmatrix}, \qquad \mathcal{G} = I_2 \otimes G = \begin{bmatrix} G & 0 \\ 0 & G \end{bmatrix}, \qquad \mathcal{J} = \begin{bmatrix} E_{\text{val}} & E_\nabla \\ E_\nabla & -E_{\text{val}} \end{bmatrix},$$

where E_{val} and $E_\nabla = I - E_{\text{val}}$ are the diagonal 0/1 matrices that select the value rows and the gradient rows of G. Then

$$\mathcal{B} = \mathcal{G}^T\mathcal{J}\mathcal{G}, \qquad \mathcal{J} = \mathcal{J}^T = \mathcal{J}^{-1}, \qquad \|G\|^2 = \|R^TA_\beta R\| \le c_1. \tag{11}$$

This is a J-factorization, as in LDL^T with $D = \pm 1$ or hyperbolic QR. G maps coefficients to values v and gradients g at the quadrature points, for both y and q. $\mathcal{J}$ is the matrix of the bilinear form

$$z^T\mathcal{B}z = \|v_y\|^2 - \|v_q\|^2 + 2\langle g_y, g_q\rangle.$$

On value rows it multiplies the y-block by $+1$ and the q-block by -1, giving $\pm R^TMR$. On gradient rows it swaps the two blocks, giving the off-diagonal $\sqrt{\beta}R^TKR$.

- $\mathcal{J}$ is a signed permutation matrix. On the device it is a single two-qubit gate acting on the qubit that labels the block (y or q) and the qubit that labels the row type (value or gradient): "if value row, flip the sign of the q-block; if gradient row, swap the blocks". Its subnormalization is 1.
- The right-hand side is $\widetilde{R}^Tf = \mathcal{G}^Tg$, where g is $\hat{y}_q$ placed in the value rows of the y-block (zero elsewhere).
- Outputs are accessed through $\mathcal{G}$ as well, since $G_{\text{val}}z_y = F_{\text{val}}y$. For example, $\langle w, y\rangle_M$ is an overlap of two states prepared with $\mathcal{G}$. So R is never applied on its own (D5).

3.5 Block encoding of the multilevel factor

A product encoding of $F_\nabla P_\ell$ loses a factor h_ℓ/h. The reason is that $\|F_\nabla\|$ is attained by oscillatory vectors and $\|P_\ell\|$ by smooth ones. Numerically, the loss reaches 148 for $L = 8$ (Table 4). The remedy is to differentiate on the coarse level and only then transfer to the fine level.

Lemma 3.6 (commuting with prolongation; cf. [8, Thm. 6.2])**.** *For each level ℓ there is an isometry T_ℓ ($T_\ell^TT_\ell = I$) with*

$$F_{\text{val}}P_\ell = T_\ell F_{\text{val},\ell}, \qquad F^{(i)}P_\ell = T_\ell F_\ell^{(i)} \quad (i = 1, \dots, d).$$

T_ℓ is element-local and the same on every coarse element. In 1D, $T_\ell = I_{2^\ell} \otimes \widehat{T}_{L-\ell}$, where $\widehat{T}_k \in \mathbb{R}^{2^{k+1}\times 2}$ maps the two weighted Gauss values of an affine function on a coarse element to its weighted values at the 2^{k+1} fine Gauss points. In d dimensions, T_ℓ is the d-fold Kronecker product.

Proof. On a coarse element, coarse functions and their partial derivatives are affine in each direction. The two Gauss points are unisolvent for affine functions, which defines $\widehat{T}_k$. Both the coarse and the fine weighted values compute the exact L^2 norm of the same affine function, because two-point Gauss quadrature is exact for quadratics. Hence $\widehat{T}_k$ is an isometry. □

For Q_1 and the gradient part, Lemma 3.6 is the commutative diagram of Deiml and Peterseim [8, Thm. 6.2]: differentiate on the coarse level, then include the coarse discontinuous space into the fine one by a map with orthonormal columns. The finite-difference analogue is [11, Lemma 4.3]. The version with values and gradients, i.e. with the mass term, is [27, Lemma 3.3]. Our contribution here is the quadrature formulation, which covers Q_k and P_k elements (Remark 3.1).

Some matrix T with $FP_\ell = TF_\ell$ always exists when F_ℓ has full column rank, e.g. $T = FP_\ell F_\ell^+$. The content of Lemma 3.6 is that T_ℓ is an isometry and element-local. This is what gives a cheap block encoding with subnormalization one. By the lemma,

$$G = \begin{bmatrix} T_1 G_1, & \dots, & T_L G_L \end{bmatrix}, \qquad G_\ell = \omega_\ell \begin{bmatrix} F_{\mathrm{val},\ell} \\ \beta^{1/4} F_\ell^{(1)} \\ \vdots \\ \beta^{1/4} F_\ell^{(d)} \end{bmatrix}, \qquad G_\ell^T G_\ell = \theta_\ell \delta_\ell^{-1} A_{\beta,\ell}. \tag{12}$$

Proposition 3.7. *G has a block encoding with subnormalization $\alpha_G \le C_d\sqrt{L}$, where C_d depends only on d.*

Sketch.

- G_ℓ has 2^d nonzeros per row and $(d+1)4^d$ per column.
- By the definition of δ_ℓ, its entries are bounded by 1, uniformly in h and β. The sparse-access encoding therefore gives $\alpha(G_\ell) \le (2^d(d+1)4^d)^{1/2}$.
- Each T_ℓ is an isometry, so it can be completed to a unitary.
- Concatenating the L levels costs a factor $\sqrt{L}$.

□

Since $\mathcal{B} = \mathcal{G}^T \mathcal{J} \mathcal{G}$, the subnormalization of $\mathcal{B}$ is $\alpha_G^2 = O(L)$. This matches the linear growth in L that Deiml and Peterseim obtain for the preconditioned stiffness factor [8, Thm. 6.3]. The sparse-access constant is pessimistic ($C_3 \approx 45$). Encoding G_ℓ as a weighted stack of Kronecker products of 1D two-term factors should give constants close to $\|G_\ell\| \le \sqrt{\theta_\ell}\, 3^{d/2}$.

From bound to measurement. The gap between the lower bound $\|G\| \approx 2$ (Table 1) and the upper bound $C_d\sqrt{L}$ matters, because α_G enters the cost squared. For $d = 3$ the two bounds differ by a factor of about 2000 in α_G^2. We therefore plan to build the block encoding of Proposition 3.7 as an explicit unitary matrix for small grids ($d = 1$ with $L \le 6$, $d = 2$ with $L \le 4$), using exactly the factors a circuit would use:

- LCUs of shifts for the 1D two-term factors, as in the example of Section 2;
- Kronecker products;
- the weighted stack over values and gradients;
- the isometries T_ℓ, completed to unitaries;
- the concatenation over levels.

For this matrix we check two things:

1. it is unitary, and its leading block equals G/α_G (correctness);
2. the value of α_G as a function of L, d and β, compared with $\|G\|$ (cost). This shows whether the $\sqrt{L}$ appears and how large C_d really is.

The effective condition number that QSVT faces is then $\kappa_\alpha = \alpha_G^2/\sigma_{\min}(\mathcal{B})$, not the ideal $\|G\|^2/\sigma_{\min}(\mathcal{B})$. **[TODO: carry out; can the $\sqrt{L}$ be avoided?]**

Theorem 3.8. *Under Assumption 3.4, QSVT applied to $\mathcal{B} = \mathcal{G}^T\mathcal{J}\mathcal{G}$ solves $\mathcal{B}z = \mathcal{G}^T g$ to accuracy ε, in the sense of* (1)*, using*

$$O\Big(\frac{C_d^2 L}{c_0}\log\frac{1}{\varepsilon}\Big)$$

calls to block encodings of G, G^T and $\mathcal{J}$. Each call costs $\mathrm{poly}(L)$ *gates. The count depends on h only through $L = \log_2 h^{-1}$, and on β only through c_0.*

Proof. By (11), Theorem 3.5 and Proposition 3.7, $\kappa_\alpha(\mathcal{B}) \le \sqrt{2}\,\alpha_G^2/c_0 \le \sqrt{2}\,C_d^2 L/c_0$. The result follows from (2). □

[TODO: End-to-end statement: preparation of g, amplitude amplification of the success probability, estimation of $\|z\|$, and readout of the QoI.]

4 The time-periodic heat equation

4.1 Problem, discretization and Fourier transform in time

We solve

$$\min_{y,u}\ \frac{1}{2}\int_0^T \|y-\hat{y}\|^2_{L^2(\Omega)}dt + \frac{\beta}{2}\int_0^T \|u\|^2_{L^2(\Omega)}dt \quad \text{s.t.} \quad \partial_t y - \Delta y = u, \quad y|_{\partial\Omega} = 0, \quad y(\cdot,0) = y(\cdot,T). \tag{13}$$

Classically, this problem is treated with multiharmonic finite elements and robust block preconditioners [12, 13].

- *Space:* Q_1 as in Section 3.
- *Time:* $N_t = 2^{L_t}$ steps of length $\tau = T/N_t$ with periodic backward Euler, $M(y_k - y_{k-1})/\tau + Ky_k = Mu_k$, with $y_0 = y_{N_t}$.

With the cyclic shift Σ and $D_t = (I-\Sigma)/\tau$, the constraint reads

$$\mathcal{K}y = \mathbb{M}u, \qquad \mathcal{K} = D_t \otimes M + I \otimes K, \qquad \mathbb{M} = I \otimes M.$$

We approximate the objective by the rectangle rule. As in (6), we eliminate u and set $q = p/(\tau\sqrt{\beta})$. This gives the symmetric system

$$\mathcal{A} = \begin{bmatrix} \mathbb{M} & \sqrt{\beta}\,\mathcal{K}^T \\ \sqrt{\beta}\,\mathcal{K} & -\mathbb{M} \end{bmatrix}, \qquad x = \begin{bmatrix} y \\ q \end{bmatrix}, \qquad f = \begin{bmatrix} \mathbb{M}\hat{y} \\ 0 \end{bmatrix}, \qquad u = \frac{q}{\sqrt{\beta}}. \tag{14}$$

D_t is circulant: $D_t = \mathcal{F}^*\Lambda\mathcal{F}$, with the unitary DFT matrix $\mathcal{F}$ and

$$\lambda_k = \frac{1-e^{-2\pi i k/N_t}}{\tau} = \frac{2}{\tau}\sin\Big(\frac{\pi k}{N_t}\Big)e^{\,i(\pi/2 - \pi k/N_t)}, \qquad \operatorname{Re}\lambda_k \ge 0. \tag{15}$$

Applying $\mathcal{F}$ to the time index block-diagonalizes $\mathcal{A}$ into the Hermitian blocks

$$\mathcal{A}_k = \begin{bmatrix} M & \sqrt{\beta}(K + \bar{\lambda}_k M) \\ \sqrt{\beta}(K + \lambda_k M) & -M \end{bmatrix}, \qquad k = 0, \dots, N_t - 1. \tag{16}$$

On a quantum computer, $\mathcal{F}$ is the QFT on the L_t time qubits, with $O(L_t^2)$ gates. It is unitary, so it does not change the condition number or the subnormalization. The analysis below uses only $\operatorname{Re}\lambda_k \ge 0$. It therefore also covers multiharmonic discretizations, where $\lambda_k = i\omega_k$.

4.2 A robust preconditioner per frequency

Theorem 4.1. *Let $r_k = 1 + \sqrt{\beta}|\lambda_k|$, $A_k = r_k M + \sqrt{\beta} K$ and $P_k = \mathrm{blkdiag}(A_k, A_k)$. Then for all h, τ, β and k,*

$$\mathrm{spec}(P_k^{-1}\mathcal{A}_k) \subset [-1, -1/\sqrt{3}] \cup [1/\sqrt{3}, 1].$$

Proof. With $X^T M X = I$ and $X^T K X = \mathrm{diag}(\mu_j)$, the pencil decouples into 2×2 pencils with eigenvalues $\pm\sqrt{1+x^2}/(1+c)$, where $x = \sqrt{\beta}|\mu_j + \lambda_k|$ and $c = \sqrt{\beta}(\mu_j + |\lambda_k|)$.

- $c \geq x$, by the triangle inequality.
- $c \leq \sqrt{2}\, x$, because $\mu_j + |\lambda_k| \leq (\mu_j + \mathrm{Re}\,\lambda_k) + |\,\mathrm{Im}\,\lambda_k| \leq \sqrt{2}|\mu_j + \lambda_k|$ when $\mathrm{Re}\,\lambda_k \geq 0$.

Hence $\sqrt{1+x^2}/(1+\sqrt{2}x) \leq \sqrt{1+x^2}/(1+c) \leq 1$. The lower bound is minimal at $x = \sqrt{2}$, where it equals $1/\sqrt{3}$. □

Remark 4.2.

- *Time domain.* In the time domain the preconditioner is $\mathrm{blkdiag}(\mathcal{P}_t, \mathcal{P}_t)$ with $\mathcal{P}_t = I \otimes (M + \sqrt{\beta}K) + \sqrt{\beta}|D_t| \otimes M$ and $|D_t| = (D_t^T D_t)^{1/2}$. It is real and symmetric, although $\mathcal{K}$ is not. $|D_t|$ is the discrete analogue of $|\partial_t|$, which underlies the $H^{1/2}$-in-time norms of parabolic theory.
- *Relation to prior work.* Theorem 4.1 is the analogue, for this discretization, of the robust multiharmonic preconditioners of [12,13]. **[TODO: compare constants and attribute.]**

4.3 Each frequency is a Poisson problem

We can write $A_k = r_k(M + \sqrt{\beta_k}K)$ with $\sqrt{\beta_k} = \sqrt{\beta}/r_k$. Each frequency is therefore the operator of Section 3 with effective parameter β_k, scaled by r_k.

- *Multilevel factor.* $R_k = [\omega_{1,k}P_1, \ldots, \omega_{L,k}P_L]$ with $\omega_{\ell,k}^2 = \theta_{\ell,k}/\delta_{\ell,k}$, where $\delta_{\ell,k} = r_k(\frac{2h_\ell}{3})^d + \sqrt{\beta}\frac{2d}{h_\ell}(\frac{2h_\ell}{3})^{d-1}$ and $\theta_{\ell,k} = \min(1, \sqrt{\beta_k}h_\ell^{-2})^s$.
- *Frequency enters only through scalars.* The prolongations P_ℓ and isometries T_ℓ are independent of k.
- *The level weights are no longer optional.* β_k decreases from β ($k = 0$) to $\beta\tau^2/(\tau + 2\sqrt{\beta})^2$ ($k = N_t/2$), so high frequencies are mass-dominated.

Factorization with a scaled reflection. Let

$$G_k = \begin{bmatrix} \sqrt{r_k}\, F_{\mathrm{val}} R_k \\ \beta^{1/4} F_\nabla R_k \end{bmatrix}, \qquad G_k^* G_k = R_k^T A_k R_k.$$

Then $\mathcal{B}_k := \widetilde{R}_k^T \mathcal{A}_k \widetilde{R}_k = \mathcal{G}_k^* \mathcal{J}_k \mathcal{G}_k$, where

$$\mathcal{J}_k = \begin{bmatrix} j_{11}E_{\mathrm{val}} & j_{12}E_{\mathrm{val}} + E_\nabla \\ j_{21}E_{\mathrm{val}} + E_\nabla & j_{22}E_{\mathrm{val}} \end{bmatrix}, \qquad J_k = \begin{bmatrix} j_{11} & j_{12} \\ j_{21} & j_{22} \end{bmatrix} = \frac{1}{r_k}\begin{bmatrix} 1 & \sqrt{\beta}\bar{\lambda}_k \\ \sqrt{\beta}\lambda_k & -1 \end{bmatrix} = s_k U_k. \quad (17)$$

On value rows, $\mathcal{J}_k$ applies the 2×2 matrix J_k across the two blocks. On gradient rows, it swaps the blocks as before. U_k is a Hermitian unitary 2×2 matrix (a reflection) with phase $\arg\lambda_k = \pi/2 - \pi k/N_t$, and

$$s_k = \frac{\sqrt{1 + \beta|\lambda_k|^2}}{1 + \sqrt{\beta}|\lambda_k|} \in \left[\tfrac{1}{\sqrt{2}}, 1\right].$$

Hence $\|\mathcal{J}_k\| \leq 1$, and $\mathcal{J}_k$ is block encoded with subnormalization 1. The weight $\sqrt{r_k}$ on the value rows is essential: without it, $\|J_k\|$ would grow like $\sqrt{\beta}/\tau$.

Theorem 4.3. *Suppose Assumption 3.4 holds for all A_k with constants c_0, c_1 independent of k. Then the nonzero eigenvalues of $\mathcal{B} = \mathrm{blkdiag}_k(\mathcal{B}_k)$ lie in $[-c_1, -c_0/\sqrt{3}] \cup [c_0/\sqrt{3}, c_1]$, and $\kappa_\alpha(\mathcal{B}) \le \sqrt{3}\,\alpha_G^2/c_0$ with $\alpha_G \le C_d\sqrt{L}$. In particular, neither N_t nor τ enters κ_α.*

Proof. Argue as for Theorem 3.5 and Proposition 3.7, frequency by frequency, with Theorem 4.1 in place of Theorem 3.2. □

QSVT applies $\mathcal{B}^+$ to all frequencies simultaneously, in superposition. Compared with the Poisson problem, the cost per solve increases only by two QFTs and the frequency-controlled scalars (Section 5).

5 Implementation

Registers.

- $d(L+1)$ qubits for quadrature rows (element index and Gauss point per direction).
- dL qubits for nodal unknowns. Level-ℓ data use the leading ℓ bits per direction. The unused index $2^\ell - 1$, a boundary node, is mapped to zero by a comparator flag.
- $\lceil \log_2 L \rceil$ qubits for the level, $\lceil \log_2(d+1) \rceil$ for the component c, one qubit b for the block, and L_t qubits for time or frequency.
- Ancillas for the block encodings and QSVT.

Coarse element indices are the leading bits of fine element indices, so $T_\ell = I \otimes \widehat{T}_{L-\ell}$ acts on trailing bits only. Unused index combinations lie in the kernel of G, which is harmless by (D4).

Circuits.

1. *1D two-term factors $F_{0,\ell}$ and $F_{1,\ell}$*: a shift by one on ℓ qubits ($O(\ell)$ gates), an LCU of two diagonal patterns, and the boundary flag. Kronecker products and a weighted stack over c then give G_ℓ.
2. *The isometry $\widehat{T}_k$*: its two orthonormal columns are the uniform vector (Hadamard gates) and a normalized discrete ramp with Gauss offsets. **[TODO: reference for linear-amplitude state preparation.]**
3. *Levels*: G_ℓ and T_ℓ controlled on the level register, followed by Hadamard gates and post-selection on the level register.
4. $\mathcal{J}$: one two-qubit gate (Poisson). For the heat equation, $\mathcal{J}_k$ needs U_k controlled on c and k, plus one ancilla rotation for s_k.
5. *QSVT phase factors* for an odd polynomial approximating $1/t$ on $[\sigma_{\min}/\alpha, 1]$, of degree $O(\kappa_\alpha \log(1/\varepsilon))$. They are computed classically, e.g. as in [7], using a lower bound for c_0.
6. *Frequency scalars (heat).* $\arg \lambda_k$ is affine in k, so it needs only single-qubit phase gates. The magnitude $|\lambda_k| = \frac{2}{\tau}\sin(\pi k/N_t)$ is an LCU of $\mathrm{diag}(e^{\pm i\pi k/N_t})$. The derived quantities $r_k^{\pm 1/2}$, $\omega_{\ell,k}$ and s_k come from QSVT polynomials or reversible arithmetic. **[TODO: compare costs.]**
7. *Right-hand side and readout.* Prepare g (weighted values of $\hat{y}$ at Gauss points, and time steps for the heat problem); for smooth $\hat{y}$, efficient circuits exist **[TODO: refs]**. Then apply $\mathcal{G}^T$, after a QFT in time for the heat problem. QoIs are overlaps $\langle \mathcal{G}g_w, \mathcal{G}z \rangle$, estimated by Hadamard tests and amplitude estimation; $\|z\|$ follows from the success probability.

Classical precomputation. All weights are in closed form. The only nontrivial precomputation is the QSVT phase factors and the constant c_0 that enters them.

6 Numerical experiments

All experiments use dense or sparse linear algebra in double precision, with scripts in the folder `preconditioning/code` of the accompanying repository:

- `qp_poisson_check.py`: Table 1, Table 4;
- `qp_bpx_weights.py`: Table 2;
- `qp_bpx_sparse.py`: Table 3, extreme eigenvalues by Lanczos with CG-based inner solves;
- `qp_heat_periodic_check.py`: Table 5.

Condition numbers on the range are ratios of the largest to the smallest nonzero eigenvalue magnitudes.

Poisson control: preconditioner and factorization. Table 1 shows:

- $\kappa(\mathcal{A})$ grows like $\sqrt{\beta}h^{-2}$;
- the interval of Theorem 3.2 is attained;
- $\kappa(\mathcal{B})$ on the range follows $\kappa(CA_\beta)$, as predicted by Theorem 3.5;
- $\|G\|$ is $O(1)$. This is the smallest possible α_G; the value achieved by the explicit construction is still to be measured (Section 3, "From bound to measurement");
- the factorization $\mathcal{B} = \mathcal{G}^T\mathcal{J}\mathcal{G}$ holds to machine precision in all cases.

d	L	β	$\kappa(\mathcal{A})$	$\lvert\operatorname{spec}(P^{-1}\mathcal{A})\rvert$	$\kappa(CA_\beta)$	$\kappa(\mathcal{B})$	$\lVert G\rVert$
1	4	1	$1.0 \cdot 10^2$	$[0.913, 1.000]$	3.45	3.49	1.86
1	6	1	$1.7 \cdot 10^3$	$[0.913, 1.000]$	4.36	4.38	2.09
1	8	1	$2.6 \cdot 10^4$	$[0.913, 1.000]$	4.98	4.99	2.23
1	8	10^{-4}	$2.6 \cdot 10^3$	$[0.708, 1.000]$	4.43	4.50	2.11
1	8	10^{-8}	$2.6 \cdot 10^1$	$[0.707, 0.999]$	7.31	8.03	2.70
2	5	1	$2.1 \cdot 10^2$	$[0.953, 1.000]$	4.06	4.04	1.75
2	5	10^{-4}	$4.0 \cdot 10^1$	$[0.707, 0.996]$	3.96	3.47	1.74
2	5	10^{-6}	$4.2 \cdot 10^0$	$[0.707, 0.981]$	6.40	7.90	2.34

Table 1: Poisson control with plain BPX ($s = 0$). $\kappa(\mathcal{B})$ is computed on the range. $\|G\|$ is a lower bound for the subnormalization α_G of any block encoding of G.

Robustness in h and β, including 3D. Tables 2 and 3 compare plain and weighted BPX.

- For fixed β, $\kappa(CA_\beta)$ is bounded in h in all dimensions. It levels off slowly (3D, $\beta = 1$: 4.2, 5.3, 6.1 for $L = 3, 4, 5$).
- Plain BPX degrades as $\beta \to 0$, up to 44.6 in 3D.
- The weights (8) with $s = 1$ keep $\kappa(CA_\beta) \le 6.2$ whenever the grid resolves β ($h \lesssim \beta^{1/4}$). In unresolved cases they approach the Jacobi constant 3^d of the mass matrix, with 20.5 as the worst value in 3D.

		β					
d, L	s	1	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
1, 10	0	5.42	5.32	5.02	4.92	7.33	9.80
1, 10	1/2	5.42	5.30	4.97	4.55	3.88	2.86
1, 10	1	5.42	5.33	6.14	5.37	4.92	3.61
2, 4	0	3.58	3.49	3.62	7.23	18.89	26.31
2, 4	1/2	3.58	3.33	2.29	3.12	9.84	14.14
2, 4	1	3.58	3.29	2.26	1.97	7.16	10.51

Table 2: $\kappa(CA_\beta)$ for the level weights (8), including the intermediate exponent $s = \frac{1}{2}$. In 2D with $L = 4$, the grid does not resolve $\beta \le 10^{-8}$.

d	L	n	s	$\beta = 1$	10^{-2}	10^{-4}	10^{-6}	10^{-8}
2	6	3969	0	4.44	4.36	4.22	7.14	11.27
2	6	3969	1	4.44	4.27	3.53	2.83	1.64
3	3	343	0	4.16	4.23	5.95	17.36	35.95
3	3	343	1	4.16	3.71	2.77	8.80	20.47
3	4	3375	0	5.30	5.31	6.41	15.75	44.55
3	4	3375	1	5.30	4.90	3.39	4.26	17.69
3	5	29791	0	6.08	6.05	6.65	14.19	–
3	5	29791	1	6.08	5.74	4.46	3.12	–

Table 3: $\kappa(CA_\beta)$ on larger grids, plain ($s = 0$) and weighted ($s = 1$) BPX, computed by Lanczos.

Subnormalization loss of the naive product. Table 4 gives the ratio $\|F_1\|\|P_\ell\|/\|F_1P_\ell\|$. A product encoding of F_1P_ℓ loses this factor. It equals $2^{L-\ell}$, up to a correction on the coarsest levels. Lemma 3.6 removes it completely: the commuted factors satisfy $\|T_\ell\|\,\|F_{1,\ell}\| = \|F_1P_\ell\|$.

ℓ	1	2	3	4	5	6	7
$\|F_1\|\|P_\ell\|/\|F_1P_\ell\|$	147.8	65.8	32.2	16.0	8.0	4.0	2.0

Table 4: Loss of the naive product encoding, $d = 1$, $L = 8$.

Time-periodic heat control. Table 5 shows:

- the global eigenvalues of $\mathcal{P}^{-1}\mathcal{A}$ lie in the interval $[1/\sqrt{3}, 1]$ of Theorem 4.1;
- per frequency, the factorization $\mathcal{B}_k = \mathcal{G}_k^*\mathcal{J}_k\mathcal{G}_k$ holds and $\|J_k\| \le 1$;
- the per-frequency BPX constants behave as in the Poisson case with effective parameters β_k.

7 Complexity and discussion

Accuracies. Let tol be the target accuracy of a QoI. It is split into three parts:

- the discretization error;
- the solver accuracy ε of (1);
- the readout error η of Section 2.

d	L	N_t	β	$\kappa(\mathcal{A})$	$\lvert\operatorname{spec}(\mathcal{P}^{-1}\mathcal{A})\rvert$
1	5	32	1	$4.1 \cdot 10^2$	$[0.714, 1.000]$
1	5	32	10^{-2}	$2.9 \cdot 10^2$	$[0.603, 0.999]$
1	5	32	10^{-4}	$4.1 \cdot 10^1$	$[0.666, 0.992]$
1	5	32	10^{-6}	$4.2 \cdot 10^0$	$[0.702, 0.990]$
2	3	16	1	$1.3 \cdot 10^1$	$[0.811, 0.999]$
2	3	16	10^{-2}	$1.2 \cdot 10^1$	$[0.669, 0.993]$
2	3	16	10^{-4}	$2.7 \cdot 10^0$	$[0.684, 0.936]$
2	3	16	10^{-6}	$4.4 \cdot 10^0$	$[0.704, 0.981]$
[TODO: per-frequency BPX and factorization checks]					

Table 5: Time-periodic heat control. The bound of Theorem 4.1 is $1/\sqrt{3} \approx 0.577$.

Each part is chosen of order tol. Up to constants, the cost of a QoI to accuracy tol is therefore (calls per solve with $\varepsilon \sim \text{tol}$) times (readout repetitions $\sim \text{tol}^{-1}$).

Poisson control. Suppose the discretization error of the QoI is $O(h^2)$, so $h \sim \text{tol}^{1/2}$.

- Classically, the cost is $O(n) = O(\text{tol}^{-d/2})$ with optimal preconditioning.
- On a quantum computer, Theorem 3.8 gives polylog(1/tol) calls per solve. Readout adds a factor $O(\text{tol}^{-1})$, for a total of $\widetilde{O}(\text{tol}^{-1})$.

An asymptotic advantage therefore requires $d = 3$ on this count. This is consistent with Deiml and Peterseim [8, Remark 6.7]. In their analysis, if the accuracy demands $h \approx \text{tol}^{1/s}$, an advantage requires $d > s$. They measure the error in the energy norm, where $s = 1$, which gives an advantage from $d = 2$. For quantities of interest with $O(h^2)$ accuracy, $s = 2$, so $d = 3$ is needed. **[TODO: check the precise statement of their Remark 6.7.]**

Heat control.

- With backward Euler, $N_t \sim \text{tol}^{-1}$ and the classical cost is $\text{tol}^{-(d/2+1)}$. On that count, a quantum advantage would appear for every d.
- This comparison is unfair. For data that are smooth in time, a multiharmonic discretization needs only log(1/tol) harmonics, and the elliptic count returns.
- The time dimension favours the quantum solver only when many harmonics are genuinely needed, i.e. for targets or forcing that are rough in time. **[TODO: quantify.]**

Limitations.

- The algorithms need fault-tolerant hardware.
- The constants (C_d, the 3^d floor, readout) matter in practice.
- State preparation for general data and the end-to-end success probability need a full analysis.
- The robustness of the multilevel factor in β is so far supported numerically, not proven.

8 Conclusions and outlook

Preconditioning for quantum linear solvers is a numerical linear algebra problem with its own rules. Respecting them leads to factorized, well-scaled preconditioned operators whose effective condition numbers are robust in the discretization parameters. Open directions:

1. a proof of β-robust multilevel bounds, and a better mass factor;
2. removing the $\sqrt{L}$ factor and computing explicit constants;
3. control constraints, where semismooth Newton or interior point methods change the $(2,2)$ block in every iteration. This links to preconditioners for interior point methods and to quantum interior point methods [21];
4. initial value problems. There D_t is not circulant. The α-circulant preconditioner [18] needs the non-unitary scaling $\Gamma_\alpha = \mathrm{diag}(\alpha^{k/N_t})$, whose condition number $1/\alpha$ enters the subnormalization. This trade-off between preconditioner quality and encoding cost has no classical analogue. Wavelet-in-time bases [23] are an alternative;
5. variable coefficients, boundary control, and non-Cartesian meshes.

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